Experiment 4: Demonstration of Network Communication Models (OSI / TCP-IP) Using a Network Simulation Tool

Aim

To demonstrate the OSI and TCP/IP network communication models using a network simulation tool (Cisco Packet Tracer).

Objective

- Understand the layers of the OSI and TCP/IP models.
- Build a small network in a simulator.
- Observe how data moves through each layer using simulation mode.

Tool Required

Cisco Packet Tracer (free for registered Networking Academy users). GNS3 or Filius can be used as alternatives.

Theory: OSI and TCP/IP Layers

OSI Layer	TCP/IP Layer	Function	Example
7 Application	Application	User services	HTTP, DNS, FTP
6 Presentation	Application	Data format, encryption	SSL/TLS, JPEG
5 Session	Application	Session control	NetBIOS, RPC
4 Transport	Transport	End-to-end delivery	TCP, UDP
3 Network	Internet	Logical addressing, routing	IP, ICMP
2 Data Link	Network Access	Frames, MAC addressing	Ethernet, Switch
1 Physical	Network Access	Bits over the medium	Cables, Hub

Network Topology Used

Two PCs connected to a switch, and a server connected to the same switch (star topology).

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0
Server0	192.168.1.100	255.255.255.0

Procedure

1. Open Packet Tracer and place 2 PCs, 1 server and 1 switch (2960) on the workspace.
2. Connect each device to the switch with copper straight-through cables.
3. Open each device, go to Desktop > IP Configuration and enter the IP address and subnet mask from the table.
4. Open the Command Prompt of PC0 and run `ping 192.168.1.11` and `ping 192.168.1.100`.
5. Switch to **Simulation Mode** and add a simple PDU (ICMP) from PC0 to Server0.

6. Click **Capture / Forward** and watch the packet move; click the packet to open the PDU Information window.
7. Examine the OSI Model tab to see the layers handled at the PC, the switch and the server.
8. Add an HTTP request from PC0 (Web Browser) to the server and compare the layers involved.

Attach screenshot here - topology with devices and cables

Attach screenshot here - ping output and PDU layer details

Observations

Device	Layers involved	Observation
PC (sender)	Layers 7 to 1	Data encapsulated: data, segment, packet, frame, bits
Switch	Layers 1 and 2	Reads destination MAC address and forwards the frame
Server (receiver)	Layers 1 to 7	Data de-encapsulated layer by layer and reply is sent

The first ping may show one 'Request timed out' because ARP resolves the MAC address first; later pings succeed with 0% loss.

Results

Communication between the devices was successful and the movement of data through the OSI and TCP/IP layers was observed in simulation mode.

Conclusion

The experiment shows how data is encapsulated at the sender and de-encapsulated at the receiver. Layered models make networks easier to design, troubleshoot and standardize.